

SOC INCIDENT RESPONSE REPORT

Project Title

SOC Home Lab: Threat Detection and Incident Response Using Wazuh SIEM

Report Information

Field	Details
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Role	Future SOC Analyst
Environment	Home Lab (Wazuh SIEM, Ubuntu Server, Windows 10, Kali Linux)
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Executive Summary

A Security Operations Center (SOC) Home Lab was deployed to simulate real-world security monitoring, threat detection, and incident response operations. The environment consisted of a Kali Linux attacker workstation, a Windows 10 endpoint monitored by Sysmon and Wazuh Agent, and an Ubuntu Server hosting the Wazuh SIEM platform.

During the assessment, multiple security events were generated and investigated, including PowerShell execution, failed authentication attempts, user account creation activity, and network reconnaissance simulations. Security telemetry was collected through Sysmon and Windows Event Logs, forwarded to Wazuh SIEM, and analyzed through the Threat Hunting module.

The objective of the exercise was to validate endpoint visibility, detection capabilities, investigation workflows, and MITRE ATT&CK mapping processes commonly used by enterprise SOC teams.

Section 1: Environment Overview

Infrastructure

System	Function	IP Address
Kali Linux	Attack Simulation Platform	192.168.207.129

System	Function	IP Address
Ubuntu Server	Wazuh SIEM Platform	192.168.207.128
Windows 10	Monitored Endpoint	192.168.207.130

Security Components

- Wazuh Manager
- Wazuh Dashboard
- Wazuh Indexer
- Wazuh Agent
- Microsoft Sysmon
- Windows Event Logging

Monitoring Objectives

- Endpoint Visibility
- Threat Detection
- Alert Investigation
- Threat Hunting
- MITRE ATT&CK Mapping
- Incident Response Validation

Section 2: Incident Identification

Security Event 1

Suspicious PowerShell Activity

A PowerShell process execution event was detected on the monitored Windows endpoint. Wazuh generated an alert indicating that a PowerShell process spawned another PowerShell instance.

Detection Source

- Sysmon Event ID 1
- Wazuh Rule ID 92027

Severity

Medium

MITRE ATT&CK

Technique:

T1059.001 – PowerShell

Tactic:

Execution

Security Impact

PowerShell is frequently leveraged by adversaries for script execution, malware delivery, credential theft, lateral movement, and post-exploitation activities.

Security Event 2

Failed Authentication Attempts

Multiple failed logon attempts were generated and analyzed through Windows Security Events.

Detection Source

Windows Security Logs

Event ID

4625

Severity

Medium

MITRE ATT&CK

Technique:

T1110 – Brute Force

Tactic:

Credential Access

Security Impact

Repeated authentication failures may indicate password guessing, brute force attempts, credential stuffing, or unauthorized access attempts.

Security Event 3

User Account Creation Activity

A user creation event was generated and monitored to validate account management visibility.

Detection Source

Windows Security Logs

Event ID

4720

Severity

Medium

MITRE ATT&CK

Technique:

T1136 – Create Account

Tactic:

Persistence

Security Impact

Unauthorized account creation can provide attackers with long-term persistence and privileged access within a compromised environment.

Security Event 4

Network Reconnaissance Simulation

Network scanning activities were conducted from the Kali Linux system using Nmap to simulate attacker reconnaissance behavior.

Tools Used

- Nmap
- TCP SYN Scan
- Aggressive Scan

MITRE ATT&CK

Technique:

T1595 – Active Scanning

Tactic:

Reconnaissance

Security Impact

Reconnaissance is often the first stage of an attack and may identify exposed services, open ports, and vulnerable systems.

Section 3: Investigation Process

Alert Triage

The cybersecurity analyst reviewed Wazuh alerts generated from endpoint telemetry and validated alert legitimacy using Sysmon and Windows Event Logs.

Threat Hunting Activities

Threat Hunting was conducted through the Wazuh Threat Hunting dashboard.

Activities included:

- Event correlation
- Process analysis
- Rule review
- Timestamp validation
- Endpoint investigation
- Security event analysis

Log Sources Reviewed

- Sysmon Operational Logs
 - Windows Security Logs
 - Windows System Logs
 - Wazuh Detection Rules
 - Wazuh Threat Hunting Events
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Section 4: Findings

The investigation confirmed successful operation of the SOC monitoring environment.

Key Findings

- Successful endpoint telemetry collection
- Centralized log aggregation
- Detection of PowerShell activity
- Monitoring of authentication events
- User account monitoring
- Threat hunting visibility
- MITRE ATT&CK correlation
- Alert investigation workflows

No evidence of malicious compromise was identified. All events originated from controlled laboratory testing conducted for educational and validation purposes.

Section 5: MITRE ATT&CK Mapping

Security Activity	Technique	Tactic
PowerShell Execution	T1059.001	Execution
Failed Logins	T1110	Credential Access
User Account Creation	T1136	Persistence
Nmap Reconnaissance	T1595	Reconnaissance

Section 6: Recommendations

Detection Engineering

- Expand Sysmon configuration coverage.
- Enable advanced process monitoring.
- Develop custom Wazuh detection rules.

Endpoint Security

- Implement PowerShell logging.
- Enable application control policies.
- Monitor administrative account activity.

Authentication Security

- Enforce Multi-Factor Authentication (MFA).
- Implement account lockout policies.
- Monitor authentication anomalies.

Threat Hunting

- Conduct regular threat hunting exercises.
- Integrate threat intelligence feeds.
- Expand ATT&CK-based detections.

Section 7: Lessons Learned

This project provided practical experience in:

- SIEM Administration
- Endpoint Monitoring
- Security Operations
- Alert Triage
- Threat Hunting
- Incident Response
- MITRE ATT&CK Analysis
- Log Correlation
- Detection Engineering

The successful deployment and operation of the SOC Home Lab demonstrates foundational skills expected of an entry-level SOC Analyst and provides hands-on experience aligned with enterprise security operations.

Conclusion

The SOC Home Lab successfully replicated a real-world security monitoring environment using Wazuh SIEM, Sysmon, Windows 10, Ubuntu Server, and Kali Linux.

The project demonstrated the complete security operations lifecycle, including event collection, detection, investigation, threat hunting, MITRE ATT&CK mapping, and incident response documentation. These activities reflect core responsibilities performed by SOC Analysts in modern Security Operations Centers.